

Arya Patil

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EDUCATION

University of Michigan — B.S., Computer Science

GPA 3.6 • May 2024

Relevant Coursework: Human-Centered Software Development, Machine Learning, Data Structures & Algorithms, Web Systems

EXPERIENCE

Broadridge — Software Engineer, Tax Servicing & AI Innovations

Newark, NJ • Jul 2025–Present

- Develop full-stack Canadian tax-form compliance features across Java, SQL, and XML on an enterprise tax-processing platform, delivering updates for multiple client accounts in parallel to meet fixed tax-year-end filing and testing deadlines.
- Diagnose off-hours production data misalignments in client-facing pipelines by tracing WebSphere application logs and underlying SQL data flows, then deploy verified fixes through the team's Jenkins pipeline with proper escalation ahead of the next business day.
- Translate client-specific regulatory requirements into scoped user stories during sprint planning and development sessions, coordinating delivery across product, QA, and client-delivery teams.

Broadridge — Career Foundations Program, Technology Track

Newark, NJ • Jul 2024–Jul 2025

- Scoped and delivered independent, stakeholder-facing projects across multiple business units — including Wealth Management, Capital Markets, Asset Management, and Enterprise Risk — on compressed two- to three-week cycles, presenting completed solutions directly to senior leadership at each close.
- Investigated recurring support-ticket patterns through interviews with developers and client support representatives, then proposed an AI knowledge-base and embedded chatbot roadmap projected to reduce inbound support volume by 25%.
- Built a Python benchmarking tool with Pandas and the Yahoo Finance API to decompose mutual fund holdings by security weighting and map them against index returns, integrating the output into a web application wealth advisors used to evaluate portfolio performance.

KEY PROJECTS

Personal Project — PokeFight, an original Pokémon-themed autobattler

Jan 2026–Present

- Designed and shipped a complete autobattler solo, from initial concept through a playable live product, establishing an original team-composition system spanning dozens of unit archetypes and cross-unit synergies.
- Ideated and implemented every unit ability and combat animation, translating source material into readable moment-to-moment gameplay and tweaking visual feedback through direct player testing to sharpen combat clarity.
- Engineered adaptive AI opponents through iterative optimization, replacing static difficulty scaling with agents that independently discover effective strategies across thousands of simulated matches.
- Balanced units and systems by combining personal playtesting, structured player feedback, and agentic simulation data, building a pipeline that turns match data into balance reports and AI-generated player discussion to surface sentiment trends alongside win rates.

Broadridge — MIPS Optimization Agent

Jun 2026–Present

- Recognized an opportunity to apply AI reasoning to a purely manual mainframe cost-optimization process, evaluating it against simpler rule-based alternatives before selecting an agent-based approach best suited to open-ended exploration.
- Orchestrated an AI agent workflow, built on Anthropic's Model Context Protocol, that runs guided mainframe job-control searches to identify the highest-cost programs and recommend fixes, translating each recommendation into direct cost savings of roughly \$1,800 per 100 MIPS reduced.

Broadridge — COBOL-to-Java Migration Agent

Jul 2026–Present

- Partnered with engineering leads to weigh migration approaches against a fixed two-month cloud deadline, then proposed agent-assisted translation to meet the accelerated timeline.
- Led an AI-assisted migration pipeline, orchestrated through Model Context Protocol tooling, that translated legacy COBOL tax-calculation programs into Java, applying test-driven development to confirm zero behavioral regressions between the original and translated systems.

Broadridge — Project Management Acceleration Through AI

Oct 2024–Apr 2025

- Identified that leadership review meetings relied on stale, manually assembled portfolio data with no centralized alerting signal, then scoped a three-part AI reporting initiative spanning dashboarding, conversational insight, and automated document generation.
- Delivered a Power BI alert dashboard, a natural-language AI chatbot built end-to-end with a Python/Flask backend and a React front end, and an automated PowerPoint generator that assembled quarterly leadership decks directly from portfolio data, cutting manual reporting preparation from days to hours.

SKILLS

Programming Python, Java, SQL, TypeScript/JavaScript, C++, C#, HTML/CSS

Design & Analysis Systems Design, Rapid Prototyping, Data-Driven Balancing, Playtesting & User Research, UX/UI Design, Technical Writing

Tools & Platforms AWS, WebSphere, Jenkins, PartyKit/Cloudflare, Power BI, Git, Jira, Confluence, Agile/Scrum